

Amit Kumar

Software Engineer | Backend Developer | Full Stack Engineer
+91 8669223920 | amitk200703@gmail.com | [LinkedIn](#) | [Github](#) | [leetcode](#) | [portfolio](#)

Profile

Software Engineering student with experience in backend development, scalable web applications, REST APIs, and distributed systems concepts. Skilled in Python, JavaScript, FastAPI, Node.js, SQL/NoSQL databases, and building production-oriented applications with focus on performance, scalability, and clean architecture.

Education

Indian Institute of Information Technology, Sri City *Andhra Pradesh, India*
B.Tech in Electronics and Communication Engineering CGPA: 7.85/10 | Expected Graduation: 2026

Experience

AI/ML Research Intern — Defence Research and Development Organisation (DRDO) May – July 2025
Visakhapatnam, Andhra Pradesh | [Kaggle](#) | [Report](#)

- Built and optimized object detection pipelines using YOLOv8/YOLOv11 on 8,000+ sonar images using PyTorch.
- Developed scalable preprocessing and evaluation workflows for model training and inference automation.
- Improved inference efficiency and reduced processing latency through pipeline optimization and debugging.
- Collaborated on experimentation, testing, and performance analysis for real-time detection systems.

Rajasthan Government School — Freelanced Project *Aug – Sept 2025*
Full-Stack Development | Rajasthan, India | [Code](#) | [Live](#)

- Designed and deployed a **scalable full-stack school management system** on AWS EC2.
- Implemented **JWT-based authentication with RBAC** for secure access control.
- Optimized backend performance using indexing and pagination, reducing response time by **15%**.
- Configured **CI/CD pipelines using GitHub Actions**.

Projects

FlowTalk: Real-Time Messaging Platform *Node.js | Express.js | Socket.io | PostgreSQL* | [Code](#)

- Planned a **real-time messaging backend** using WebSockets (Socket.io).
- Implemented **event-driven architecture** for real-time message delivery and updates.
- Built **JWT authentication with refresh tokens** for scalable session management.
- Developed **WebRTC signaling server** and message persistence using PostgreSQL.
- Created system to handle concurrent real-time users with efficient message delivery and persistence.

Scalable Financial Processing Platform *Python | FastAPI | LangChain | PostgreSQL | Redis | Docker* | [Code](#) | [Demo](#)

- Built backend services and RESTful APIs using FastAPI for financial workflow automation and reporting systems.
- Designed modular service-oriented architecture with PostgreSQL and Redis for scalable data processing and caching.
- Implemented asynchronous task workflows, API integrations, and automated reconciliation pipelines.
- Improved system maintainability through containerized deployment using Docker and structured backend architecture.

Meal-Monkey *MongoDB | Express.js | React.js | Node.js | Redis | Docker* | [Code](#) | [Live](#)

- Developed a scalable full-stack food delivery platform using MERN stack with multi-role authentication.
- Built REST APIs for order management, cart handling, and real-time order tracking.
- Improved backend performance using Redis caching, indexing, and pagination techniques.
- Tested and debugged backend APIs using Postman and structured exception handling practices.
- Implemented JWT authentication, RBAC, and Dockerized deployment for secure and maintainable services.

Technical Skills

- Languages:** Python, JavaScript, TypeScript, C++
- Backend & Frameworks:** Node.js, Express.js, FastAPI, REST APIs
- Frontend:** React.js, HTML5, CSS3
- Databases:** PostgreSQL, MySQL, MongoDB, Redis, SQL
- Tools & Cloud:** Docker, Git, Linux, AWS, CI/CD, Github Actions
- Testing & debugging:** Postman, JEST(basic), API testing
- Core CS:** Data Structures & Algorithms, OOP, DBMS, OS, Computer Networks, Backend Architecture Basics

Achievements

- Solved 400+ DSA problems with strong focus on optimization and edge cases
- Built and deployed multiple scalable backend applications using Docker, Redis, PostgreSQL, and cloud services.